LipScribe: Teaching a Computer to Read Lips and Listen at the Same Time

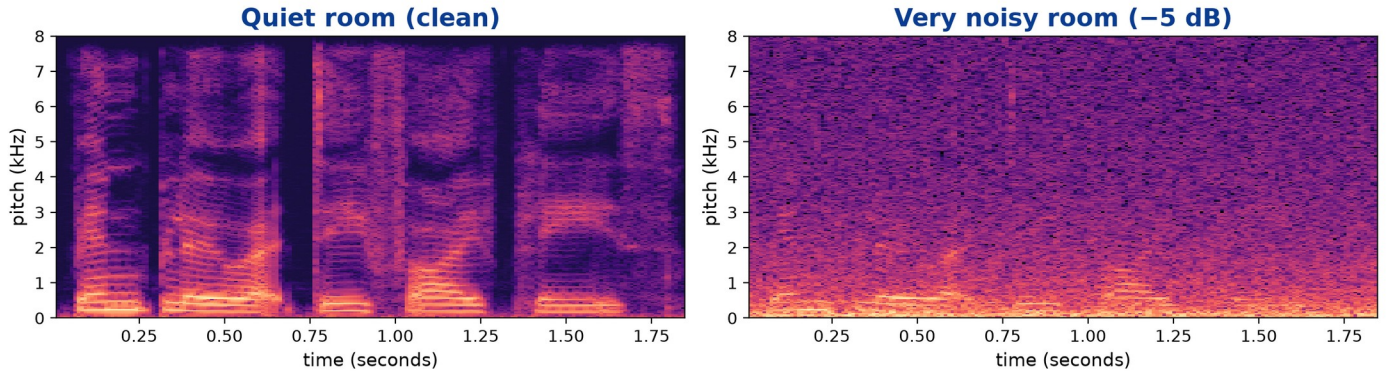
Progress so far (Weeks 1–4) · Computer Vision and Deep Learning · Group 4

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12 November 2026

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The problem: captions break when a room gets loud



A ‘sound picture’ of the same sentence. The bright stripes are the voice; on the right, noise almost hides them.

Who this helps: people who are deaf or hard of hearing, and anyone captioning a lecture, market, taxi park or video call where the audio is poor.

1 Sound gets buried

Noise drowns out the voice, so a listen-only system starts guessing.

2 Lips don’t change

Noise can’t touch what the camera sees. The mouth moves the same way.

3 Our idea: use both

Listen and lip-read together, and lean on the lips when it gets loud.

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Where we are, in four numbers

1.0%

words wrong in a quiet room

about 1 word in 100

0.7%

with our grammar checker on

fewer than 1 word in 100

−43%

fewer errors in the noisiest room

5.3% vs 9.5% for listen-only

1.9%

with the camera switched off

still as good as listen-only

In short: all planned models are built and tested. Our final design, Model P, makes the fewest mistakes overall, cuts errors in noise, and keeps working if the camera or microphone fails.

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Four words you'll hear in this talk

M Model

A program that learns a skill from examples instead of following hand-written rules. We show it thousands of videos with their correct sentences, and it slowly learns the pattern.

% Word error rate (WER)

Out of every 100 words spoken, how many the model gets wrong. 2% means 2 mistakes per 100 words. Lower is better.

dB Noise level (dB)

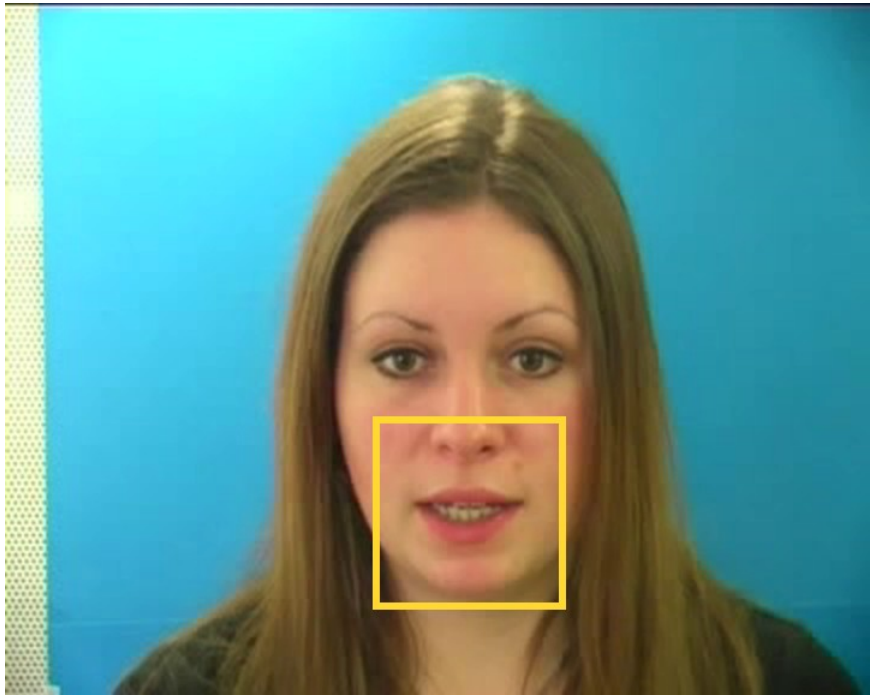
How loud the background is compared to the voice. "Quiet" has no noise; at −5 dB the noise is about 3× stronger than the speaker.

? Unseen speakers

We test only on people the model never saw while learning, so the score shows how it would do on a brand-new person.

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Our data: 33 people, 32,890 short video clips



A real GRID clip. The yellow box is the part we keep.

Every clip is a 3-second, six-word sentence, for example:

“bin blue at d five please”

26
people to learn from

3
people to tune on

4
new people to test on

Quality checks first. An automatic audit of all 34,000 files found one speaker with no video and 110 broken clips. We removed them and kept 96.7% of the data.

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What the computer actually looks at



Six moments from the sentence “bin blue at d five please”: the lip shape changes with each sound.

1 Find the face

A face-finding tool marks 40 points around the lips in every frame.

2 Cut out the mouth

Crop a steady square around the lips, grey, 96×96 pixels.

3 Keep the timing

Never drop a frame: a missed frame is filled in from its neighbours.

4 Line up the sound

Shift the audio so each sound lands on the right lip movement.

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How LipScribe turns a video into a sentence

Lip video

75 mouth pictures

Sound

voice recording

Eye part

learns lip shapes

Ear part

learns sounds

Combiner

merges what the eyes and ears found, moment by moment

Reader + grammar check

spells out letters, then snaps to the nearest valid sentence

“bin blue at d five please”

The key design question is the combiner: how should the system mix the two senses? We tried a simple version first (baselines), then built a smarter one (Model P).

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Progress against our plan

M1	M2	M3–M4	M5	M6–M7	M8–M9
Set up computers and check the data	Prepare all 32,890 clips	Build the three simple models	Build the smart combiner (Model P)	Fine-tune and test thoroughly	Live caption app and demo
Done · 29 Oct	Done · 3 Nov	Done · 4 Nov	Done · 5 Nov	In progress · due 22 Nov	Started · due 29 Nov

Team of six, one shared machine: every model was trained on our Alienware workstation (RTX 4080 SUPER graphics card), and every result was re-checked there on 12 November.

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Step 1: three simple models to beat

B1 Ears only

Listens to the sound.

2.0% wrong in a quiet room

9.5% wrong in the noisiest room

Great when it's quiet, but errors grow almost 5× in noise.

B2 Eyes only

Reads the lips, no sound at all.

23% wrong, in every room

Noise has zero effect, but some sounds look the same on the lips (b, p, m).

B3 Eyes + ears, glued

Sticks both together in one step.

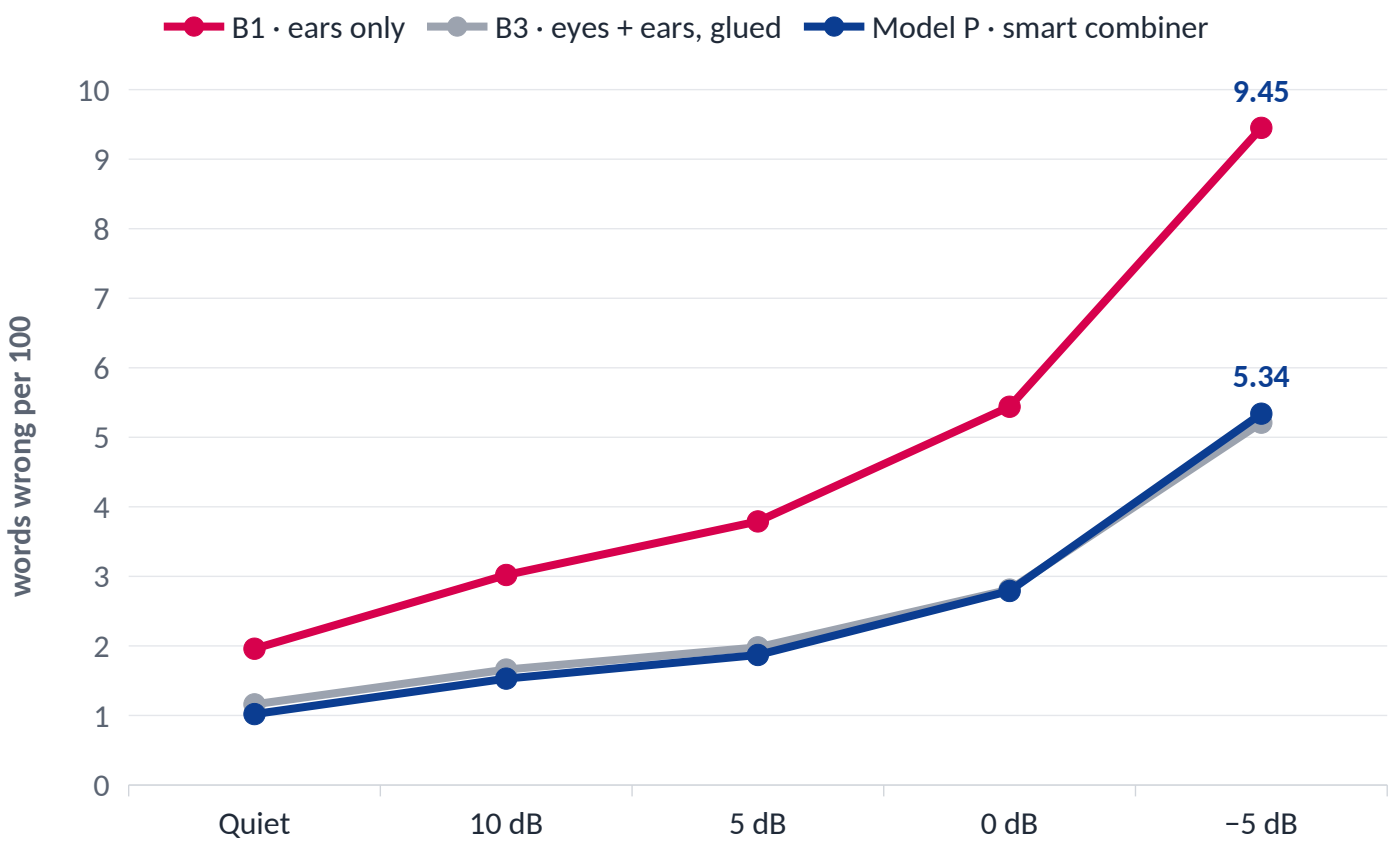
1.2% wrong in a quiet room

5.2% wrong in the noisiest room

Better than B1 everywhere, but it falls apart if the camera or mic stops.

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Using both senses beats listening alone in every room



Quiet room

Model P: 1.0 wrong per 100 words, the best of all our models (ears only: 2.0).

Noisiest room

43% fewer errors than ears only (5.3 vs 9.5). Glued B3 is level here (5.2).

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Why we upgraded the ‘eyes’ to ResNet-18

Old eyes: simple 3D-CNN

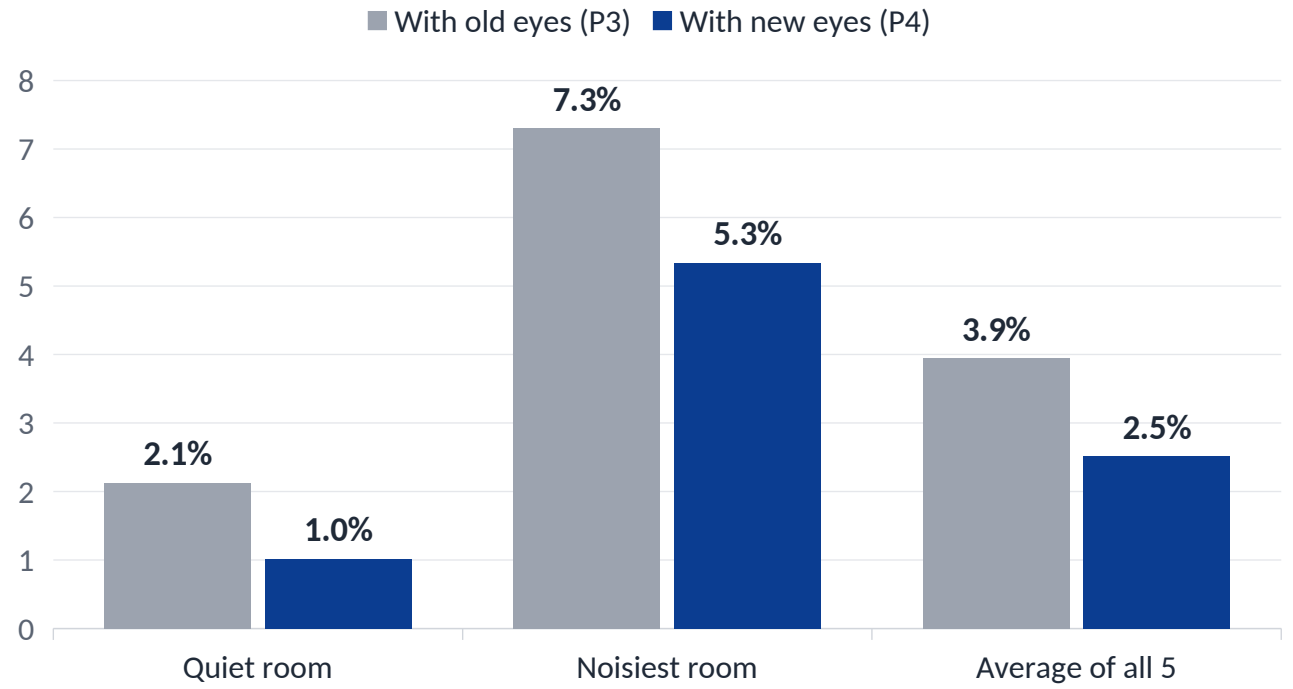
Only 3 layers deep.

Tied to exact pixel positions, so a new face shape or lighting threw it off: 13% to 34% wrong depending on the person.

New eyes: ResNet-18

17 layers with ‘shortcuts’ that make deep learning stable.

A proven design, but it learned everything from our own GRID videos.



Result: half the mistakes in a quiet room (2.1% → 1.0%), and the combiner trusts the lips nearly twice as much in noise.

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Borrowed design, our own learning

1 Copied the blueprint

Like building from a proven house plan, but laying every brick ourselves. ResNet-18 comes from published lip-reading research; it learned everything from our GRID videos, starting from random.

Technical: adopted architecture, trained from scratch

2 Reused our own experience

Like hiring someone who already knows half the job. Model P starts with the trained eyes (B2) and ears (B1), then learns how to combine them.

Technical: transfer learning between our models

3 Took GRID skills to films

Later in this talk: for the film clips, the model starts from what it learned on GRID instead of from zero.

Technical: transfer learning to a new dataset

In one sentence: we adopted the ResNet-18 lip-reading architecture and trained it from scratch on GRID; we then used transfer learning to reuse those GRID-trained parts in Model P and in the film-clip experiments.

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Step 2: Model P, a smarter way to combine the senses

1

The senses check each other

Ears can glance at nearby lip movements to fill in a word lost in noise; eyes can use the sound to tell look-alike lip shapes apart (b, p, m).

Technical name: cross-modal attention

2

A trust dial for every moment

25 times a second, the model sets a dial between ‘trust the ears’ and ‘trust the lips’. Loud room: turn towards the lips.

Technical name: gated fusion

3

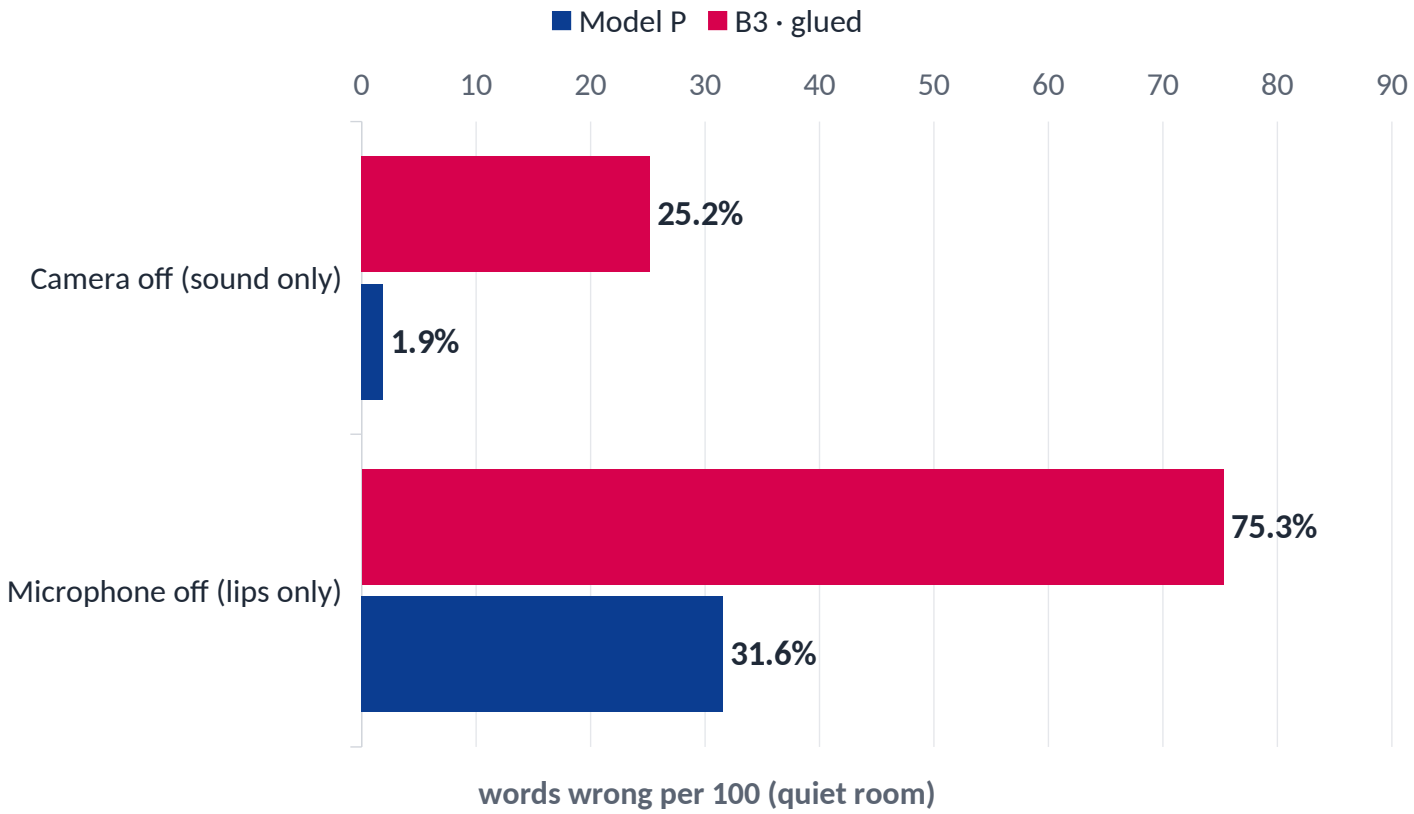
A stand-in when a sense is missing

If the camera loses the face or the mic cuts out, a learned ‘nothing here’ signal fills the gap so the model knows to rely on the other sense.

Technical name: missing-modality embedding

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If the camera or mic fails, Model P keeps going



Camera off

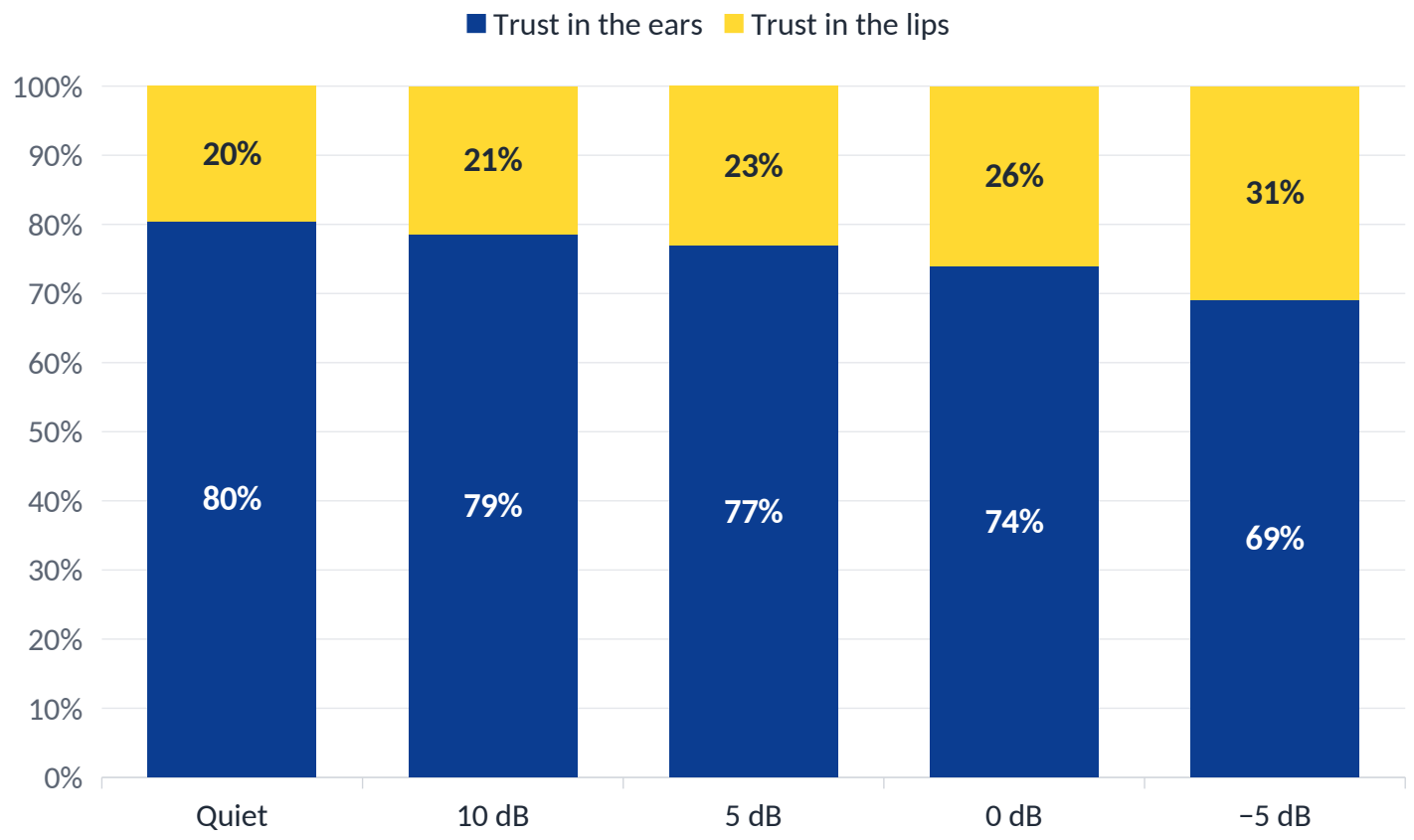
Model P: 1.9% wrong, as good as the ears-only model (2.0%). Glued B3: 25%.

Microphone off

Model P: 32% wrong, as good as lips-only. Glued B3: 75%, nearly useless.

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As the room gets louder, Model P trusts the lips more



20% → 31%

share of trust given to the lips, quiet room → noisiest room

Why it matters

The model learned by itself to lean on the lips when the sound gets bad: exactly the behaviour we designed it for.

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From letter guesses to words: greedy vs grammar

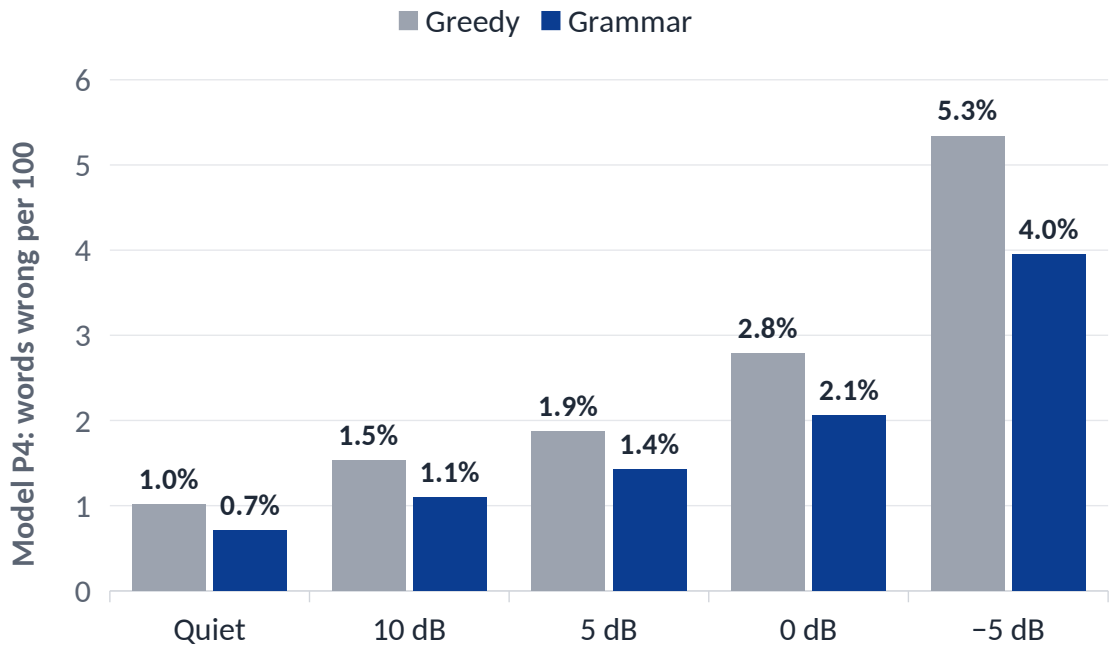
Greedy: best letter each moment

Merge repeats, drop the gaps. Fast, but one weak letter spells a wrong word.

Grammar: only real GRID sentences

Every sentence is command · colour · preposition · letter · digit · adverb, from 51 words. The decoder picks the allowed sentence that best fits the guesses.

Real fixes (0 dB):
place white by x five **snow** → **now**
lay red by k **ninhe** please → **nine**



555 test sentences fixed, none made worse. Films have no fixed pattern, so there we use the open-ended version: a word list + which words usually follow which.

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Problems we hit, and how we fixed them

1 Sound ran ahead of the lips

The audio files had their silence cut off, so speech started up to 1.5 s early. We re-aligned every clip using its timing file.

2 Missing and broken files

One speaker had no video and 110 clips wouldn't open. An automatic audit found and removed them.

3 The model 'cheated'

In a practice run all 8 sentences began "bin blue at f", so it guessed the same answer for all. Mixed examples forced it to really read lips.

4 Eyes didn't work on new people

The old eye part ranged from 13% to 34% wrong depending on the person. Fixed by switching to ResNet-18.

5 Gluing made quiet rooms worse

Training both senses from zero let the weak lip side add noise. Fixed by starting from the trained B1 and B2.

6 Crashes when a sense was missing

Blanking a whole sense made the maths divide by zero. The learned 'stand-in' signal fixed it.

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Teaching Model P film dialogue



Real mouth crops our script found in films, 7 moments per line, with the subtitle that labels them.

The film data

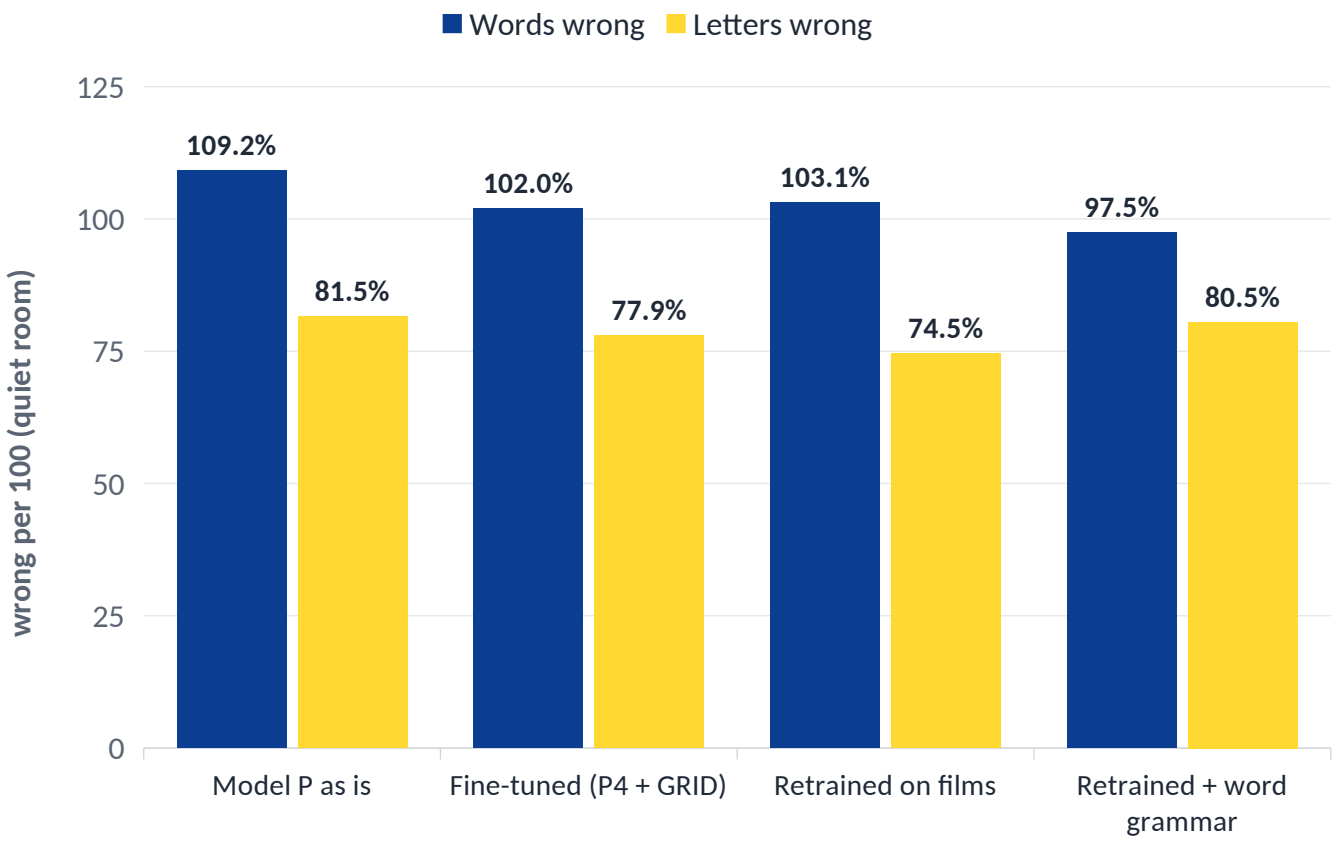
305 clips from 13 titles: 9.7 min, 651 words. 3 films kept aside for the test.

Three things we tried

1. Fine-tune Model P, GRID mixed in
2. Retrain it the P4 way, films only: 26 settings tested, best one × 3 runs × 150 rounds
3. Word grammar: only real words, from 9,053 in the training films’ subtitles

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Results on three films it never saw



Better

Letters: 81.5% → 74.5% wrong. Grammar: first time under 100% words wrong.

Still not working

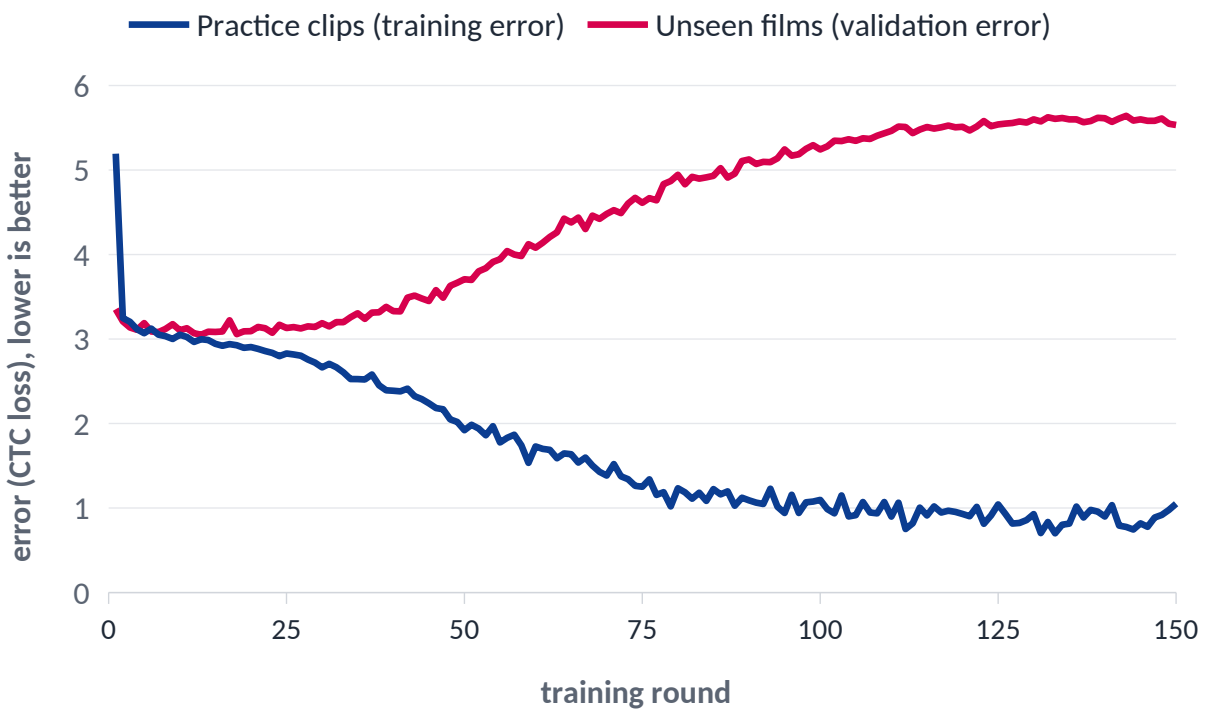
About 1 word in 30 right, short ones: you, and, what. It also forgot GRID (0.8% → 115%).

Why

9.7 minutes of film. Research systems use hundreds of hours; ~7× more is being collected.

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Watching it train: practice vs. unseen films



Best round: 46 (75.3% of letters wrong on unseen films). After that it memorises.

Alienware · training log (real)

round	practice error	unseen error	letters wrong
1	5.19	3.35	100.0%
10	3.05	3.10	100.0%
30	2.66	3.19	86.4%
46	2.18	3.58	75.3% ★
100	1.10	5.24	78.3%
150	1.05	5.53	78.5%

150 rounds · ~4 s each · RTX 4080 SUPER

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What our model wrote, scene by scene



Subtitle (The Drama)

“you know like uh”

Model P as is

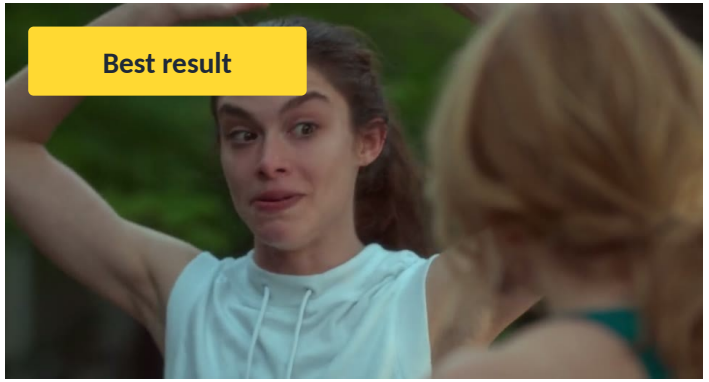
“bn ne now”

Retrained on films

“you nt a ne”

Retrained + word grammar

“you dont and”



Subtitle (Little Women)

“i used to be”

Model P as is

“lian whe win b eight sgon”

Retrained on films

“wha ma ne”

Retrained + word grammar

“i am”



Subtitle (Little Women)

“oh i thought you were a novelist not a screenwriter”

Model P as is

“lay reten bi v iro soon”

Retrained on films

“iond tao u maes srs sne te”

Retrained + word grammar

“and to me sense”

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What's next

✓ Just finished

Better sound features: ears-only errors on the tuning speakers dropped from 7.7% to 6.9%. To be merged.

First version of the caption app is in the code base.

M6 By 22 November

Retrain Model P with the new sound features.

Fine-tune settings and finish the full test report.

M8 By 29 November

Live captions from a webcam and microphone, powered by Model P.

Final testing and the demo.

Double-checked on the Alienware (12 Nov): all 9 trained models are saved, the 'best model' file is byte-for-byte the P4 model, and the test scores in this deck match the saved results.

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